internal to specific tools, never appear in a `.chiplet` file

1 KiCad PCB internal nm

signed integers

`export_chiplet.cpp`

2 PCB-bbox-corner

used for dies and `io_pads`

3 PCB-bbox-center

used for the interposer

intermediate `.chiplet`

`_metadata.finalize_required: true`
readers must refuse to load it

4 HYP absolute

metres, KiCad y-down
before conversion

`hyp_to_gds.py` input parsing

5 GDS absolute

micrometres, y-up cartesian

`hyp_to_gds.py --update-chiplet-file`

the only place that owns the interposer GDS bbox;
strips the `_metadata` block

visible in the schema

7 GDS-bbox-center

the interposer's own
position convention

6 GDS-bbox-corner

`CANONICAL`
origin at the lower-left of
the interposer GDS bbox

=

8 Interposer-local-corner

the schema's intent,
formalised as identical to 6

`(x, y, z) µm →`
`(x/1000, z/1000, -y/1000) mm`

9 Chiplet Studio 3D world

OpenGL, millimetres, y-up,
z out of the screen